A commutator estimate for the Golden–Thompson gap in qubit chains

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18 October 2026

Abstract

For a finite qubit chain with bounded nearest-neighbor interactions, split the interaction sum into its odd and even bond layers A and B. We compare the logarithmic trace gap $F(\beta) = \log\{\tau(e^{\beta A}e^{\beta B})/\tau(e^{\beta(A+B)})\}$ with $\mathcal{E} = \tau([A,B]^*[A,B])$, where τ is the normalized trace. If every bond has operator norm at most J, then $\beta^4\mathcal{E}/48 \le F(\beta) \le \beta^4\mathcal{E}/16$ for $0 \le \beta J \le 1/16000$. The constants and interval are independent of chain length and remain valid when local commutator contributions nearly cancel. The proof retains two occurrences of the full commutator in an ordered-cumulant expansion before applying local estimates. A spanning-tree bound and a matrix row-and-column estimate then control the full Taylor remainder relative to $\mathcal{E}$. The result gives a one-step partition-function certificate and a converse on the stated high-temperature interval. An exactly solvable four-site example illustrates the role of cancellation.

Keywords. Golden–Thompson inequality; quantum spin chain; commutator; ordered cumulant; product formula.

1 The trace gap and the question of uniformity

For two Hermitian matrices, the Golden–Thompson inequality compares the trace of the exponential of their sum with the trace of the product of their exponentials [1, 2]. When the matrices are sums of local interactions, this comparison also measures a bias in a split approximation to a quantum partition function. The product formula of Trotter and subsequent time-slice analyses place such approximations in a well-established setting [3, 4]. We ask how the one-step logarithmic bias depends on the noncommutation of two alternating bond layers, with a temperature scale determined by the individual bonds.

Consider the Hilbert space $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$, where $N \ge 1$, and Hermitian operators h_i, $1 \le i < N$, supported on sites $\{i, i+1\}$. An operator supported on a subset of those two sites is allowed, as is a scalar component. Assume that

$$\|h_i\| \le J, \qquad J \ge 0, \tag{1.1}$$

where the norm is the operator norm. Define

$$A = \sum_{i \text{ odd}} h_i, \qquad B = \sum_{i \text{ even}} h_i, \qquad H = A + B, \qquad \tau(X) = 2^{-N}\operatorname{Tr} X. \tag{1.2}$$

The bonds within each layer have disjoint supports. The interactions may vary from bond to bond; no translation invariance is assumed.

The quantities to be compared are

$$C = [A,B], \qquad \mathcal{E} = \tau(C^*C), \qquad F(\beta) = \log\frac{\tau(e^{\beta A}e^{\beta B})}{\tau(e^{\beta H})}. \tag{1.3}$$

Here $\mathcal{E}^{1/2}$ is the Hilbert–Schmidt norm defined using the normalized trace. Both traces in (1.3) are positive for real β. For the numerator, this follows from

$$\operatorname{Tr}(e^{\beta A}e^{\beta B}) = \operatorname{Tr}(e^{\beta B/2}e^{\beta A}e^{\beta B/2}) > 0.$$

The Golden–Thompson inequality gives $F(\beta) \geq 0$.

For a fixed pair A, B, expansion at zero yields

$$F(\beta) = \frac{\beta^4}{24}\mathcal{E} + O_{A,B}(\beta^5). \tag{1.4}$$

The fourth-order coefficient is established finite-order trace algebra; for example, Tao gives its explicit commutator-square form [5]. Earlier trace-correction analyses also explain why the trace has improved order relative to the unsymmetrized operator product [6, 7]. Formula (1.4) alone leaves open how small β must be when the matrices vary.

Two uniformities are needed for the present question. The sums A and B grow with the chain, whereas the bound J describes one local interaction. In addition, $\mathcal{E}$ can be much smaller than its extensive scale NJ^4, either because noncommutation is confined to a few sites or because different local contributions cancel. Even an extensive remainder estimate

$$\left|F(\beta) - \beta^4\mathcal{E}/24\right| \leq KN(\beta J)^5$$

would have relative size at most $K(\beta J)NJ^4/\mathcal{E}$ compared with $\beta^4\mathcal{E}$. It would therefore not give a common interval when $\mathcal{E}/(NJ^4)$ tends to zero. We prove a bound whose remainder retains the factor $\mathcal{E}$ itself.

Theorem 1.1. *Under* (1.1)–(1.3),

$$\frac{\beta^4\mathcal{E}}{48} \leq F(\beta) \leq \frac{\beta^4\mathcal{E}}{16} \qquad \textit{if} \qquad 0 \leq \beta J \leq \frac{1}{16000}. \tag{1.5}$$

The constants are independent of N and of the particular interactions. If $J = 0$ or $\mathcal{E} = 0$, then $F(\beta) = 0$ for every real β.

No lower bound on the commutator energy per site is required. The interval in Theorem 1.1 is deliberately conservative. Its purpose is to give a single explicit local scale on which both directions hold, including arbitrarily small nonzero $\mathcal{E}$.

1.1 Relation to earlier estimates

Fye's time-slice formulation contains exactly the expression in (1.3). In his notation, set $M = 2$, $H_1 = -A$, $H_2 = -B$, $L = 1$, and $\Delta\tau = \beta$. If ΔF_{Fye} denotes his difference between exact and approximate thermodynamic free energies, then

$$F(\beta) = \beta\,\Delta F_{\mathrm{Fye}}.$$

His equations (2.21)–(2.24) use trace cyclicity to replace the two-factor product by a symmetric product. This identity applies to arbitrary complex Hermitian matrices. Section IV of that paper develops the local, thermodynamic size-scaling argument by neglecting boundary terms relative to volume, including a coefficient-by-coefficient discussion of higher orders [4]. Those results establish important antecedents for the present problem. They do not give a finite-chain summed remainder proportional to the full $\mathcal{E}$ when that quantity is arbitrarily small.

Other quantitative comparisons control different forms of the defect. Carlen and Lieb bound the unlogged Golden–Thompson difference by an explicit weighted trace expression [8]. Childs and collaborators give imaginary-time product-formula estimates involving nested

operator commutators and global operator-norm scales [9]. Trace cyclicity identifies their one-step symmetric-product trace with our numerator, but the passage to a bound of the form (1.5) still requires local control relative to $\mathcal{E}$. Connected high-temperature expansions supply local convergence and extensive estimates in considerably broader settings [10]. The issue addressed here is the relative size of the remainder near the zero set of the full layer commutator.

The proof uses established ordered-cumulant and dependency-graph methods [11, 12]. In particular, the spanning-tree estimate used below, including its ordered noncommutative form, is prior work. We include its finite combinatorial proof to make the argument self-contained. The additional estimate comes from extracting two copies of the global C before taking absolute values and then combining equal local Pauli strings. This produces the squared coefficient sum $\mathcal{E}$, preserving cancellation that a sum of separate local squares would discard.

2 Cancellation and the direct consequences

A four-site example shows why the full commutator is the relevant quantity. Let X_j, Y_j, Z_j denote the Pauli matrices at site j, with identity operators on the other sites. For $J > 0$ and $0 \le \varepsilon \le 1$, take

$$h_1 = JZ_2, \qquad h_2 = \frac{J}{2}(X_2X_3 + Y_2Y_3), \qquad h_3 = (1-\varepsilon)JZ_3. \tag{2.1}$$

Each bond has norm at most J. The first and third bonds use the one-site components permitted in the statement.

Write $c_1 = [h_1, h_2]$ and $c_2 = [h_3, h_2]$. Direct Pauli multiplication gives

$$c_1 = iJ^2(Y_2X_3 - X_2Y_3), \qquad c_2 = -(1-\varepsilon)c_1.$$

Thus $C = c_1 + c_2 = \varepsilon c_1$, and orthogonality of distinct Pauli strings gives

$$\mathcal{E} = 2\varepsilon^2 J^4, \tag{2.2}$$

$$\tau(c_1^* c_1) + \tau(c_2^* c_2) = 2J^4\{1 + (1-\varepsilon)^2\}. \tag{2.3}$$

At $\varepsilon = 0$, the two layers commute although both local commutators are nonzero. Near that point, a bound expressed through (2.3) misses the smallness retained by (2.2).

The traces can also be calculated exactly. Sites 1 and 4 are spectators. On the central pair, H/J has eigenvalues $\pm(2-\varepsilon)$ and $\pm\sqrt{1+\varepsilon^2}$. The latter pair comes from the subspace spanned by $|01\rangle, |10\rangle$, where $A/J = \varepsilon Z$ and $B/J = X$. With $\theta = \beta J$, it follows that

$$\tau(e^{\beta A}e^{\beta B}) = \frac{\cosh((2-\varepsilon)\theta) + \cosh(\varepsilon\theta)\cosh\theta}{2}, \tag{2.4}$$

$$\tau(e^{\beta H}) = \frac{\cosh((2-\varepsilon)\theta) + \cosh(\theta\sqrt{1+\varepsilon^2})}{2}. \tag{2.5}$$

Consequently $F(\beta) = \varepsilon^2\theta^4/12 + O_\varepsilon(\theta^6)$. Figure 1 places the exact logarithmic ratio within the interval supplied by Theorem 1.1. It illustrates the information retained by the theorem; the proof below treats every chain in the stated class.

2.1 A partition-function certificate and a converse

Let $Z = \operatorname{Tr} e^{\beta H}$ and $Z_1 = \operatorname{Tr}(e^{\beta A}e^{\beta B})$. Since normalized-trace factors cancel, Theorem 1.1 gives

$$Z_1 e^{-\beta^4\mathcal{E}/16} \le Z \le Z_1 e^{-\beta^4\mathcal{E}/48}. \tag{2.6}$$

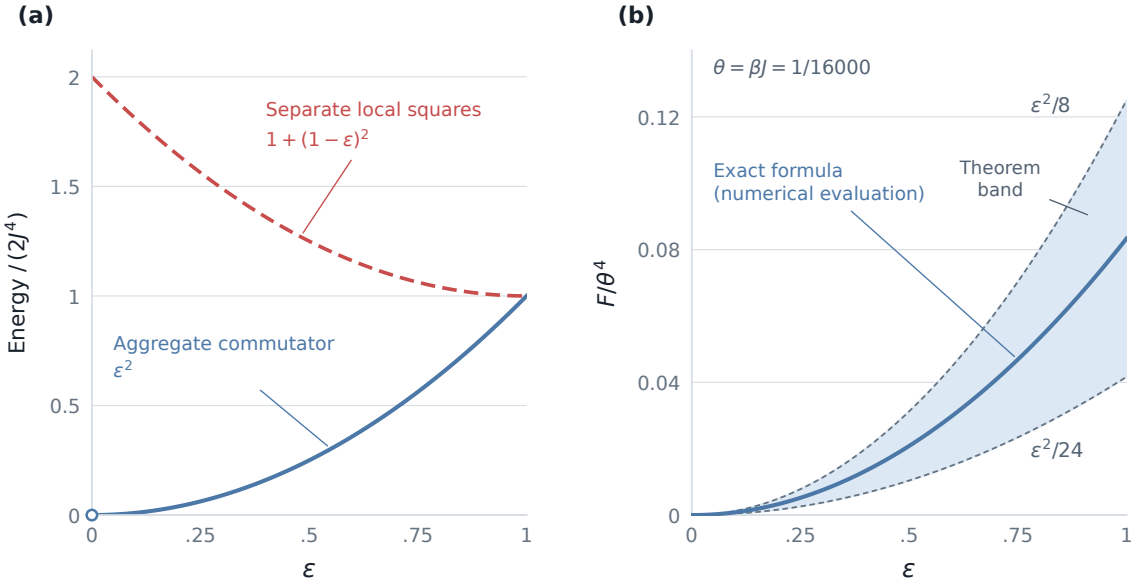


Figure 1: Cancellation in the family (2.1). Panel (a) compares $\mathcal{E}/(2J^4) = \varepsilon^2$ with the separate local squares $L/(2J^4) = 1 + (1-\varepsilon)^2$, where L is the left-hand side of (2.3). At $\varepsilon = 0$, the full commutator cancels exactly. Panel (b) shows the exact formula for F/θ^4, evaluated at $\theta = 1/16000$, within the theorem band $\varepsilon^2/24 \le F/\theta^4 \le \varepsilon^2/8$. The quartic coefficient $\varepsilon^2/12$ is visually indistinguishable from the plotted curve and is retained in the accompanying data. The shading represents theorem bounds; the plotted values are high-precision evaluations of exact formulas.

For the physical Hamiltonian $-H$, define the exact and split free-energy densities by $f = -\log Z/(\beta N)$ and $f_1 = -\log Z_1/(\beta N)$. For positive β in the stated interval,

$$\frac{\beta^3}{48}\frac{\mathcal{E}}{N} \le f - f_1 = \frac{F(\beta)}{\beta N} \le \frac{\beta^3}{16}\frac{\mathcal{E}}{N}. \tag{2.7}$$

The lower bound also gives the converse

$$\mathcal{E} \le \frac{48}{\beta^4} F(\beta). \tag{2.8}$$

Thus a small scalar gap controls aggregate noncommutation in the normalized Hilbert–Schmidt norm on this interval.

Uniform constants do not imply that the total gap is independent of N. For example, the connected chain $h_i = JZ_iZ_{i+1}$ for odd i and $h_i = JX_iX_{i+1}$ for even i has $\mathcal{E} = 4(N-2)J^4$ when $N \ge 2$, since the neighboring-bond commutators are distinct orthogonal Pauli strings. Its lower bound grows linearly with the chain length. Equations (2.6)–(2.8) concern a single product at the specified inverse temperature. They do not give an error bound for $(e^{\beta A/r}e^{\beta B/r})^r$, a real-time unitary simulation, or a sampling algorithm.

3 Extracting the two commutators

The logarithm is useful because it subtracts products of disconnected contributions. Ordered cumulants express that subtraction while keeping the order of noncommuting operators. Their adjacent-swap identity will allow us to retain the full C until both copies needed for $\mathcal{E}$ are present.

3.1 Ordered trace cumulants

For a list $X_1, \dots, X_n$, let $\mathcal{P}([n])$ denote the set of partitions of the labeled positions $1, \dots, n$. Define

$$\kappa(X_1, \dots, X_n) = \sum_{\pi \in \mathcal{P}([n])} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{S \in \pi} \tau\left(\prod_{j \in S}^{\nearrow} X_j \right). \tag{3.1}$$

The arrow means that positions within each block retain their original order. The products over distinct blocks are products of complex numbers. This distinction allows the partition formula to be used even when the X_j do not commute.

The cumulant is multilinear. It is also the coefficient of $t_1 \cdots t_n$ in

$$\log \tau(e^{t_1 X_1} \cdots e^{t_n X_n})$$

near the origin. To see this, expand the logarithm: the coefficient with each variable present once groups its positions into nonempty blocks, with coefficient $(-1)^{|\pi|-1}(|\pi| - 1)!$.

Cyclically rotating the entire list leaves κ unchanged, because the subword in each block is then cyclically rotated inside its trace. Taking adjoints gives

$$\overline{\kappa(X_1, \dots, X_n)} = \kappa(X_n^*, \dots, X_1^*). \tag{3.2}$$

Lemma 3.1 (Adjacent-swap identity)**.** *For two adjacent entries and arbitrary lists U, V,*

$$\kappa(U, A, B, V) - \kappa(U, B, A, V) = \kappa(U, [A, B], V). \tag{3.3}$$

The right-hand list has one fewer entry.

Proof. In (3.1), first consider partitions that place the two adjacent positions in different blocks. Interchanging those positions pairs equal terms in the two sums. The order of all other entries in each block is unchanged, so these terms cancel.

For partitions that place both positions in the same block, the difference of its moments replaces the adjacent product AB by $AB - BA$. Merge the two positions into one position carrying $[A, B]$. This is a bijection with the partitions of the shorter list and preserves the number of blocks. The partition coefficient is therefore unchanged, giving (3.3). □

Because $C^* = -C$, (3.2) and cyclic invariance imply, for any word W in Hermitian letters A, B,

$$\operatorname{Re} \kappa(C, W) = \frac{1}{2}\{\kappa(C, W) - \kappa(C, W^{\mathrm{rev}})\}. \tag{3.4}$$

Sorting W and its reversal to the same word, while leaving the initial C fixed, expresses this difference through adjacent swaps. Lemma 3.1 supplies a second full commutator in every resulting term. This operation precedes any expansion into individual bonds.

3.2 The coefficient weight

Write $f_n = [z^n]F(z)$, initially using the analytic logarithms near zero. If $\operatorname{sort} w$ puts all A's before all B's, multilinearity gives

$$f_n = \frac{1}{n!} \sum_{w \in \{A,B\}^n} \{\kappa(\operatorname{sort} w) - \kappa(w)\}. \tag{3.5}$$

For the first logarithm, words with a letters A and b letters B occur $\binom{n}{a}$ times, so their factor is $1/(a!b!)$, as required by the two exponentials. For the second logarithm, expand each entry of $\kappa(H, \dots, H)/n!$.

The coefficient f_n is real, so we may take real parts term by term in (3.5). Fix a word with a letters A, $b = n - a$ letters B, and inversion count $I(w)$. Sorting it requires $I(w)$ adjacent swaps. Each resulting difference contains one C. Rotate that C to the first position and apply (3.4). The remaining word contains $a-1$ copies of A and $b-1$ copies of B. Its inversion count and the inversion count of its reversal sum to $(a-1)(b-1)$. Hence each real first-swap term has a representation by cumulants containing two C's, with total absolute coefficient at most $(a-1)(b-1)/2$.

The average inversion count among words with fixed a, b is $ab/2$: reversal pairs the count I with $ab - I$. Therefore the total absolute coefficient in this two-commutator representation is at most

$$\frac{1}{4n!}\sum_{a=0}^{n}\binom{n}{a}a(a-1)b(b-1) = \frac{2^{n-6}}{(n-4)!}, \qquad n \geq 4. \tag{3.6}$$

To see the equality, choose two distinct ordered positions for A and two for B; each of the other $n-4$ positions can carry either letter. The sum before division is $n!2^{n-4}/(n-4)!$.

Every resulting cumulant has $m+2$ entries, consisting of two C's and $m = n-4$ entries from $\{A, B\}$. When fewer than two copies of either letter are available, the real difference vanishes. In particular, the Taylor coefficients below order four are zero. We now need a local bound for these two-commutator cumulants that preserves the cancellation inside C.

4 Local cumulant bounds and the full commutator energy

4.1 A spanning-tree estimate

A support graph has one vertex for each occurrence in a list of local operators, including repeated occurrences, and joins two vertices when their supports intersect. We may add edges to this graph. For a graph G, let $\mathrm{ST}(G)$ be its number of spanning trees, with $\mathrm{ST}(G) = 0$ if it is disconnected and $\mathrm{ST}(G) = 1$ for one vertex.

Lemma 4.1 (Spanning-tree bound)**.** *If G contains the support graph of $X_1, \ldots, X_k$, then*

$$|\kappa(X_1, \ldots, X_k)| \leq 2^{k-1}\,\mathrm{ST}(G)\prod_{j=1}^{k}\|X_j\|\,. \tag{4.1}$$

This is the individual-cumulant estimate of Féray, Méliot and Nikeghbali; their ordered noncommutative formulation applies here [12, arXiv v4, Sections 9.3–9.4 and 11.1.1]. We give the argument to identify exactly where locality enters.

Proof. For a subset S of occurrences, put $m(S) = \tau(\prod_{j\in S}^{\nearrow} X_j)$. If $G[S]$ has several connected components, the operator groups belonging to different components act on disjoint tensor factors. They commute, and normalized trace factorizes, so $m(S)$ is the product of the moments on those components. Moreover, $|m(S)| \leq \prod_{j\in S}\|X_j\|$.

Refine each partition π in (3.1) by splitting its blocks into the connected components induced by G. Call the resulting partition ρ. All partitions with this connected refinement have the same moment product. Form the quotient graph G/ρ by contracting each block of ρ, discarding internal edges and retaining distinct parallel edges.

The partitions π that refine to ρ correspond to partitions of the quotient vertices into independent sets. Their aggregate coefficient is consequently $[x]\chi_{G/\rho}(x)$, the coefficient of x in the chromatic polynomial. Indeed, a partition into r nonempty independent color classes contributes the falling factorial $(x)_r$, whose linear coefficient is $(-1)^{r-1}(r-1)!$.

The absolute value of this linear coefficient is at most $\mathrm{ST}(G/\rho)$. One can obtain this familiar bound from deletion–contraction: for a connected simple graph the sign of the linear

coefficient is $(-1)^{v-1}$, and its absolute value obeys the same deletion–contraction recurrence as spanning-tree counts, with the appropriate zero term when deletion disconnects the graph. Parallel edges leave the chromatic polynomial unchanged and can only increase the tree count.

It remains to sum over connected partitions. Each induced connected graph $G[S]$ has at least one spanning tree, whence

$$\sum_{\rho} \mathrm{ST}(G/\rho) \le \sum_{\rho} \mathrm{ST}(G/\rho) \prod_{S\in\rho} \mathrm{ST}(G[S]) = 2^{k-1}\,\mathrm{ST}(G). \tag{4.2}$$

The equality has a direct counting interpretation. Choose a spanning tree of G and mark any subset of its $k-1$ edges as cuts. Removing those edges gives the blocks of ρ, with a tree inside each block; the cut edges form a tree on the quotient. Conversely, these internal and quotient trees reconstruct the original tree and its marked cuts. Multiplying the moment bounds now proves (4.1). The same reasoning gives zero when G is disconnected. □

4.2 Combining equal Pauli strings

The operator $D = iC$ is Hermitian. Expand it in the standard nonidentity Pauli strings P, namely tensor products of I, X, Y, Z with at least one nonidentity factor:

$$D = \sum_{P} d_P P, \qquad d_P \in \mathbb{R}, \qquad \tau(PQ) = \delta_{P,Q}, \qquad \sum_{P} d_P^2 = \mathcal{E}. \tag{4.3}$$

Only neighboring bonds from different layers can fail to commute. Their union contains at most three consecutive sites, so every P appearing in (4.3) has a smallest containing interval of length at most three. The identity coefficient is zero because $\tau(C) = 0$.

Equal strings in this expansion are combined before estimation. Thus d_P is the total coefficient of P, including contributions from every neighboring-bond commutator. In the example of Section 2, it already includes the cancellation by the factor ε.

Lemma 4.2 (Two commutator entries)**.** *For any ordering of two entries equal to C and $m \ge 0$ entries chosen from A, B,*

$$|\kappa(C, \ldots, C, \ldots)| \le 315\, 2^{m+1}(m+2)^m 4^m J^m \mathcal{E}. \tag{4.4}$$

Here the notation denotes the whole prescribed list; the two C's may occupy any two positions.

Proof. Expand the m ordinary entries into their bond terms and use $C = -i\sum_P d_P P$ for the two other entries. For each local list, apply Lemma 4.1, using intersection of containing intervals as the graph. This graph may have more edges than the exact support graph, which is allowed. By Cayley's formula, there are $(m+2)^m$ labeled trees on the $m+2$ positions. Write the spanning-tree count as a sum of edge-intersection indicators over these trees.

Fix one such tree T and root it at the first marked Pauli string P. An ordinary bond child has at most four choices of left endpoint that intersect its parent's interval, whose length is at most three. Its norm is at most J. A marked Pauli-string child Q has at most five choices of leftmost site, and at most $4^3 - 1 = 63$ nonidentity patterns at each such site. There is one free marked child and m ordinary children. Summing from the root outwards therefore bounds the total weight, for a fixed P, by

$$M = 315\, 4^m J^m. \tag{4.5}$$

The count deliberately includes extra possibilities, so it also covers shorter supports, gaps inside a three-site interval, boundary sites and the parity restrictions on bonds.

More explicitly, let $K_T(P, Q) \ge 0$ be the sum over the ordinary bond choices of their norm products times the indicators of all tree-edge intersections. The preceding count

says $\sup_P \sum_Q K_T(P,Q) \le M$. Rooting the same tree at the other marked vertex gives $\sup_Q \sum_P K_T(P,Q) \le M$. Both estimates are needed; symmetry of the two marked positions is not assumed.

The matrix bound from row and column sums, or Schur's test, now gives

$$\sum_{P,Q} |d_P| K_T(P,Q) |d_Q| \le M \sum_P d_P^2 = M\mathcal{E}.$$

For example, this follows directly by applying $2|d_P d_Q| \le d_P^2 + d_Q^2$ and then the two sum bounds. Multiplying by the factor 2^{m+1} from Lemma 4.1 and the number $(m+2)^m$ of trees proves (4.4). □

Combining (3.6) with (4.4), for $n = m+4$, gives

$$|f_{m+4}| \le \frac{315}{2}\, 16^m \frac{(m+2)^m}{m!} J^m \mathcal{E}. \tag{4.6}$$

For $m \ge 1$, the elementary inequalities $m! \ge (m/e)^m$ and $(1+2/m)^m \le e^2$ imply

$$|f_{m+4}| \le K(RJ)^m \mathcal{E}, \qquad K = \frac{315e^2}{2}, \qquad R = 16e. \tag{4.7}$$

This controls every higher coefficient relative to the same $\mathcal{E}$. Its constants are too large to use efficiently for the fifth coefficient, so we calculate that coefficient separately before summing the remainder.

5 The low orders and the uniform remainder

5.1 The fourth and fifth coefficients

Let a_n denote the coefficient of β^n in $\tau(e^{\beta A}e^{\beta B}) - \tau(e^{\beta H})$. The coefficients below degree four vanish by cyclicity. At degree four,

$$a_4 = \frac{\tau(A^2B^2) - \tau(ABAB)}{12} = \frac{\mathcal{E}}{24},$$

since expansion of $C^*C = -C^2$ gives $\mathcal{E} = 2\{\tau(A^2B^2) - \tau(ABAB)\}$.

At degree five, group words cyclically according to whether they contain three A's or three B's. Words with three A's and two B's have two cyclic classes, represented by A^3B^2 and A^2BAB, each of size five. The product exponential contributes $\tau(A^3B^2)/(3!2!)$, whereas the sum exponential contributes $5\{\tau(A^3B^2) + \tau(A^2BAB)\}/5!$. The analogous calculation with A, B exchanged gives

$$a_5 = \frac{1}{24}\tau(A^3B^2 + A^2B^3 - A^2BAB - ABAB^2) = \frac{1}{24}\tau(HC^*C). \tag{5.1}$$

The last equality follows by expanding $-H(AB-BA)^2$ and collecting its cyclic words. Terms with only one occurrence of a letter cancel.

Because both traces equal $1 + \beta\tau(H) + O(\beta^2)$, passage to their ratio and logarithm leaves a_4 unchanged and replaces a_5 by $a_5 - \tau(H)a_4$. Thus

$$f_4 = \frac{\mathcal{E}}{24}, \qquad f_5 = \frac{\tau(HC^*C) - \tau(H)\mathcal{E}}{24}. \tag{5.2}$$

The centering term is essential when scalar bond components are present. The fifth coefficient need not vanish.

Lemma 5.1. *The fifth coefficient satisfies* $|f_5| \le 5J\mathcal{E}$.

Proof. Put $H_0 = H - \tau(H)I$. Since $D = iC$,

$$24 f_5 = \tau(H_0 D^2).$$

Consider the Hilbert space of operators with inner product $\langle U, V\rangle = \tau(U^* V)$, and its subspace spanned by all nonidentity Pauli strings whose containing interval has length at most three. It contains D. Compress left multiplication by H_0 to this subspace. The resulting matrix is Hermitian.

For Pauli strings P, Q, a Pauli summand R of H_0 contributes to $\tau(PH_0Q)$ only when Q equals RP up to a phase. If the smallest interval containing P is $[a, b]$, a bond term h_i can contribute only when

$$i \in [a - 3, b + 2].$$

Indeed, outside that range every nonidentity site of R is too far from P for the product to have a containing interval of length at most three. There are at most eight such bonds because $b - a \le 2$.

Each bond has at most fifteen nonidentity two-site Pauli coefficients, and each coefficient has absolute value at most $\|h_i\| \le J$. Centering has removed the scalar terms. The absolute row sums of the compressed multiplication matrix are therefore at most $120J$; Hermiticity gives the same column bound. Its operator norm is at most $120J$, so

$$|\tau(H_0 D^2)| = |\langle D, H_0 D\rangle| \le 120J\,\tau(D^2) = 120J\mathcal{E}.$$

Divide by 24. □

5.2 Summation and the temperature interval

For a fixed finite chain, F is real analytic on the whole real axis because its two traces are positive there. The coefficient estimate (4.7) gives a Taylor series convergent on $|\beta|J < 1/R$. It agrees with F near zero and hence throughout this connected real interval by uniqueness of real-analytic continuation. Thus the coefficient calculation supplies a bound for the actual logarithmic ratio, without taking a thermodynamic limit.

Proposition 5.2 (Relative remainder)**.** *Let* $J > 0$*, set* $\theta = \beta J$*, and use* K, R *from* (4.7)*. For* $0 \le \theta < 1/R$*,*

$$\left| F(\beta) - \frac{\beta^4 \mathcal{E}}{24} \right| \le \beta^4 \mathcal{E} \left(5\theta + \frac{K(R\theta)^2}{1 - R\theta} \right). \tag{5.3}$$

Proof. The term of degree five is bounded by $5\beta^5 J\mathcal{E}$, and (4.7) bounds the terms of degrees $m + 4$, $m \ge 2$. Sum the resulting geometric series. □

Proof of Theorem 1.1. Assume first that $J > 0$ and $\mathcal{E} > 0$. If $0 \le \theta \le 1/16000$, the elementary inequality $e < 3$ gives $R\theta < 3/1000$ and $K < 2835/2$. The parenthesis in (5.3) is at most

$$\frac{5}{16000} + \frac{(2835/2)(9/10^6)}{997/1000} = \frac{41821}{3190400} < \frac{1}{48}. \tag{5.4}$$

Combining this with the coefficient $1/24$ gives the lower coefficient $1/48$ and the upper coefficient $1/16$ in (1.5).

If $\mathcal{E} = 0$, faithfulness of the normalized trace gives $C = 0$; then A and B commute and $e^{\beta A} e^{\beta B} = e^{\beta H}$. If $J = 0$, all bonds vanish. These cases give the asserted exact zero gap. □

6 Scope of the estimate

The relative remainder in Proposition 5.2 is the part of the argument that turns the fixed-pair expansion into a uniform comparison. Its dependence on $\mathcal{E}$ follows from the order in which the proof performs its operations: the two commutators are extracted in the logarithmic expansion, equal Pauli strings are combined, and only then are local absolute bounds applied. This explains why the conclusion remains valid near the cancelling family of Section 2.

The explicit constants come from support counts and tree estimates chosen for uniformity. They do not identify a useful lower-temperature range or an optimal constant. The result supplies the one-step certificate (2.6), the free-energy-density bound (2.7) and the converse (2.8) for finite open qubit chains. Extending any of these conclusions to repeated products or a wider class of systems requires a corresponding estimate for that setting.

Reproducibility. The accompanying source package contains the LaTeX manuscript, bibliography, figure source, and data calculated from (2.4)–(2.5). High-precision arithmetic is used because the trace gap on the proved interval can be much smaller than ordinary floating-point resolution. The mathematical proof is independent of these calculations.

AI assistance. AI tools assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program.

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